

RESEARCH PROJECT REPORT

FORECAST-DRIVEN ROBUST PORTFOLIO OPTIMIZATION UNDER MACROECONOMIC UNCERTAINTY

(An Empirical Study Using Economic Policy Uncertainty Index and Convex Optimization)

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ABSTRACT

This study evaluates the predictive power of Economic Policy Uncertainty (EPU) and develops an end-to-end operational research framework translating uncertainty forecasts into optimal asset allocation decisions. Utilizing monthly U.S. data from January 1990 to December 2025 (420 observations) sourced from the FRED database, we systematically evaluate forecasting architectures across econometric and machine learning paradigms. Out-of-sample forecasting results over the test set (January 2022 to December 2025) demonstrate that the parsimonious ARIMA(1,1,1) model serves as the champion architecture, achieving the lowest Root Mean Squared Error (RMSE) of 45.57 and a Mean Absolute Percentage Error (MAPE) of 14.45%. Pairwise predictive accuracy is further validated via the Diebold-Mariano test.

To bridge the gap between economic forecasting and mathematical programming, we introduce a continuous Z-score risk transformation that maps EPU expectations into dynamic parameter adjustments within a convex portfolio optimization engine executed via CVXPY. Under rigorous out-of-sample backtesting over the turbulent 2022–2025 market contraction cycle, and net of 10 basis points transaction costs, the proposed EPU-Robust strategy achieves an annualized Sharpe ratio of 0.56. This performance marks a substantial improvement of 19.1% over the traditional Markowitz mean-variance strategy (0.47) and 55.6% over the naive equal-weight baseline (0.36), effectively overcoming the empirical estimation errors that typically challenge optimized portfolios. Furthermore, the framework demonstrates robust tail-risk management, restricting the maximum drawdown to -11.5% compared to -14.2% for the traditional mean-variance benchmark. The empirical results confirm that incorporating forward-looking macroeconomic uncertainty signals into constrained convex optimization can systematically suppress drawdowns and generate sustainable risk-adjusted economic alpha.

Keywords: Economic Policy Uncertainty, Time Series Forecasting, ARIMA, Portfolio Optimization, CVXPY, Operational Research, Asset Allocation.

1. INTRODUCTION:

Concerns about policy uncertainty have intensified in the wake of Global Financial Crisis, the COVID-19 pandemic and the post-2022 inflation surge. They have repeatedly demonstrated that unexpected policy shifts can trigger extreme market volatility, erode investor confidence, and disrupt asset price dynamics.

To quantify this phenomenon, Baker, Bloom & Davis (2016) developed the Economic Policy Uncertainty (EPU) index, which captures policy-related economic uncertainty by tracking newspaper coverage, and economic forecaster disagreements. The EPU index has since become a standard proxy for macroeconomic risk in empirical finance and has been shown to predict stock market volatility, bond yields, and gold prices.

In spite of growing evidence that EPU contains valuable predictive information for asset returns, most existing studies focus exclusively on forecasting EPU itself rather than translating those forecasts into actionable investment decisions. This gap between forecasting and decision-making represents an underexplored area in operational research.

1.1. Problem Statement:

Traditional portfolio optimization frameworks, most notably Markowitz's mean-variance model, rely exclusively on historical return distributions under the implicit assumption of stationarity that is, the statistical properties governing future asset returns will mirror those observed in the past. Although computationally tractable, this backward-looking paradigm is subject to two fundamental theoretical limitations.

Markowitz's (1952) mean-variance framework operates under the implicit assumption that the statistical properties of future returns will mirror those observed historically. DeMiguel, Garlappi and Uppal (2009) demonstrated across seven empirical datasets that none of 14 mean-variance extensions consistently outperforms the naive $1/N$ portfolio in out-of-sample Sharpe ratio, attributing this failure to estimation error amplification precisely the limitation this study addresses by incorporating a forward-looking uncertainty signal.

Han and Li (2023) the most directly related prior work demonstrated that integrating EPU forecasts into portfolio construction via the Black-Litterman model materially improves out-of-sample portfolio performance. The present research extends this framework by: (a) evaluating a substantially wider model space across nine forecasting architectures; (b) employing a convex robust optimization formulation rather than the Bayesian Black-Litterman approach; and (c) conducting out-of-sample evaluation over the genuinely turbulent 2022–2025 inflationary period.

These limitations have significant practical implications: portfolios optimized under tranquil market conditions frequently exhibit substantial performance deterioration during periods of acute financial stress, when uncertainty regimes undergo abrupt transitions. A growing body of literature has consequently advocated for the systematic integration of forward-looking

uncertainty forecasts into the portfolio construction process as a means of improving out-of-sample robustness across heterogeneous economic environments.

1.2.Research Objectives:

This study aims to bridge the gap between economic forecasting and operational research by developing an end-to-end framework that transforms EPU forecasts into optimal portfolio decisions. The specific objectives are:

- To conduct a comprehensive comparative evaluation of forecasting model specifications spanning classical econometric and modern machine learning architecture for the US Economic Policy Uncertainty Index.
- To develop a theoretically grounded transformation framework converting EPU forecasts into quantitative risk scores suitable for integration into portfolio optimization.
- To formulate and implement an EPU-driven robust convex portfolio optimization model with uncertainty-adjusted regularization constraints.
- To evaluate the proposed framework through rigorous out-of-sample backtesting across the high-uncertainty period 2022–2025, benchmarking against equal-weight and traditional mean-variance strategies.

1.3.Research Questions:

RQ1: Can parsimonious econometric time-series models achieve superior out-of-sample forecasting accuracy relative to complex machine learning and deep learning architectures for the EPU Index?

RQ2: How can EPU forecasts be systematically transformed into actionable risk signals that improve dynamic asset allocation decisions across distinct macroeconomic regimes?

RQ3: Does the EPU-robust optimization framework achieve statistically and economically meaningful risk-adjusted performance improvements over traditional mean-variance and naive benchmark strategies?

1.4.Contributions:

This research contributes to three streams of literature. First, it provides a comprehensive nine-model comparative evaluation of EPU forecasting methodologies, extending the model space considered by Han and Li (2023) beyond the CEEMDAN-GRU to include a wider range of classical and ensemble approaches. Second, it introduces a novel Z-score risk transformation framework bridging EPU forecasting and convex portfolio optimization, providing a transparent and theoretically motivated uncertainty-adjustment mechanism. Third, it empirically validates over a period of genuine macroeconomic stress (2022–2025) the proposition that forward-looking uncertainty integration materially enhances out-of-sample portfolio performance.

2. Literature Review:

2.1. Economic Policy Uncertainty and Financial Markets:

Baker, Bloom and Davis (2016) developed the EPU Index as a composite of three empirical components: newspaper article frequency mentioning economic policy uncertainty (validated against manual readings of 12,000 articles), the number of federal tax code provisions scheduled to expire, and the dispersion of professional economic forecasts. Their U.S. index spikes around tight presidential elections, the Gulf War, the 9/11 attacks, the failure of Lehman Brothers, and the 2011 debt ceiling dispute events that correspond intuitively with elevated investor uncertainty. Baker et al. (2016) further demonstrate using firm-level data that EPU is associated with greater stock price volatility and reduced investment and employment in policy-sensitive sectors.

The theoretical underpinning for EPU's financial market effects derives substantially from Bloom's (2009) real-options framework, which models how time-varying second-moment uncertainty shocks generate 'wait-and-see' dynamics in firm investment. Bloom (2009) finds that a simulated macro uncertainty shock calibrated to historical spikes produces a rapid, short-lived drop and rebound in output and employment of approximately 1% to 1.5%, establishing that second-moment (uncertainty) shocks can generate recessions without requiring first-moment (productivity) shocks.

At the asset pricing level, Pastor and Veronesi (2013) develop a general equilibrium model showing that political uncertainty commands a risk premium, making stocks more volatile and more correlated especially in weak economic conditions. Their model implies that policy uncertainty reduces the implicit put protection government provides to the market, directly raising equity risk premia. Liu and Zhang (2015) subsequently confirmed empirically that EPU significantly increases S&P 500 realized volatility both in-sample and out-of-sample, with its marginal effect of approximately 0.03–0.04% per standard deviation increase providing important predictive value for portfolio risk management. Brogaard and Detzel (2015) extended this evidence to the cross-section, documenting that EPU positively forecasts excess market returns consistent with EPU being a priced risk factor with a negative risk premium across the Fama-French 25 size-momentum portfolios.

The cross-asset implications of EPU are directly relevant to portfolio construction. Ma, Wang and He (2022) document dynamic and frequency-dependent spillover characteristics between EPU and stock markets across G7 countries, demonstrating that EPU-induced correlation shifts across asset classes create time-varying diversification opportunities that static covariance matrices fail to capture. This finding directly motivates the dynamic rebalancing mechanism central to this research.

2.2. EPU and Portfolio Optimization:

Han and Li (2023) investigated the impact of global EPU on portfolio optimization via a Black-Litterman framework, applying a hybrid CEEMDAN-GRU model to forecast EPU and XGBoost to forecast asset returns. Their empirical results demonstrate that the EPU-integrated approach outperforms Markowitz mean-variance, minimum-variance, equal-weight, and risk-parity strategies across six investment periods providing the most direct motivation for the present study's forecast-integration approach. The present research extends Han and Li (2023) in three respects: (a) a wider nine-model forecasting evaluation that includes classical approaches the authors omit; (b) a convex robust optimization formulation rather than the Black-Litterman model; and (c) explicit transaction cost incorporation in the backtesting framework.

Fabozzi, Kolm, Pachamanova and Focardi (2007) define robust optimization as the systematic incorporation of parameter uncertainty into the mathematical formulation, demonstrating that the approach preserves the quadratic programming structure whilst improving out-of-sample performance during adverse conditions. The size of the uncertainty set, as Fabozzi et al. (2007) note, 'controls the aversion to the uncertainty that comes from the estimation error' — a parameter this study anchors empirically to the EPU Z-score rather than setting heuristically. DeMiguel, Garlappi and Uppal (2009) establish a critical benchmark: the expected improvement from any forward-looking approach must overcome the severe estimation error that causes even sophisticated mean-variance models to underperform $1/N$. The EPU-robust framework's 55.6% Sharpe ratio improvement over the equal-weight baseline demonstrated in Section 4 must be understood in this context.

2.3. Time-Series Forecasting for EPU:

The choice of forecasting methodology for the EPU Index is far from obvious given the index's non-stationary, fat-tailed distributional characteristics. The Box-Jenkins (1976) ARIMA framework has dominated macroeconomic forecasting for decades due to its theoretical grounding in stationary stochastic processes, its parsimony, and its interpretable structure. The ARIMA model captures the autoregressive persistence, moving-average shock dynamics, and integrated non-stationarity that characterize monthly macroeconomic series.

Siami-Namini and Namin (2018) provide a direct empirical comparison of ARIMA versus LSTM for economic and financial time-series forecasting, finding that LSTM outperforms ARIMA on average across standard datasets — with average error reductions of 84–87% cited in their study. However, their advantage is concentrated in long-horizon prediction tasks and in environments with sufficient training data to calibrate the network's gate mechanisms reliably. For EPU forecasting specifically, the limited sample size (348 monthly training observations), the non-stationarity induced by structural breaks at crisis events, and the high signal-to-noise ratio all challenge deep learning approaches that require large, stationary datasets to generalize reliably out-of-sample.

Gradient boosting approaches, including XGBoost (Chen and Guestrin, 2016) — have demonstrated competitive performance in financial forecasting by capturing nonlinear relationships and interaction effects that linear models systematically miss. Chen and Guestrin (2016) introduce XGBoost as a scalable tree-boosting system with regularization controls that substantially reduce overfitting relative to standard gradient boosting. The walk-forward cross-validation procedure employed for XGBoost in this study is designed specifically to prevent look-ahead bias in non-stationary environments, addressing the fundamental limitation identified by Siarni-Namini and Namin (2018) for static train-test splits.

3. Data and Methodology:

3.1.Data Description:

The raw dataset initially contains 497 monthly observations spanning from 1985 to 2026. After eliminating temporal misalignments, constructing historical lag features, and restricting the research horizon to a stable macro regime (January 1990 – December 2025), the final clean research universe consists of exactly 420 observations. The empirical analysis employs monthly time-series data spanning January 1990 to December 2025, yielding 420 observations. The primary variable is the U.S. Economic Policy Uncertainty Index (USEPUINDXM) from the FRED database, corresponding to Baker, Bloom and Davis's (2016) construction. The investable asset universe comprises five ETFs: SPY (S&P 500 equity), QQQ (NASDAQ-100 technology equity), TLT (20+ Year U.S. Treasury bonds), LQD (investment-grade corporate bonds), and GLD (gold bullion) spanning equity, fixed-income, and commodity exposures.

Variable	Description	Frequency	Source
EPU (USEPUINDXM)	U.S. Economic Policy Uncertainty Index — Baker et al. (2016)	Monthly	FRED
VIX (VIXCLS)	CBOE Volatility Index (implied equity volatility)	Monthly	FRED
UNRATE	U.S. Civilian Unemployment Rate (%)	Monthly	FRED
FEDFUNDS	Federal Funds Effective Rate (%)	Monthly	FRED
GS10	10-Year Treasury Constant Maturity Rate (%)	Monthly	FRED
TB3MS	3-Month Treasury Bill Rate (%)	Monthly	FRED
CPIAUCSL	Consumer Price Index, All Urban Consumers	Monthly	FRED

INDPRO	Industrial Production Index	Monthly	FRED
UMCSENT	University of Michigan Consumer Sentiment	Monthly	FRED
YieldSpread	Term spread: GS10 – TB3MS (engineered)	Monthly	Computed

Table 1: Variable Description and Data Sources (1990–2025)

The data is divided into three periods following standard time-series validation practice:

Period	Time Range	Months	Purpose
Training	1990-01 to 2018-12	348	Model estimation
Validation	2019-01 to 2021-12	36	Hyperparameter tuning
Test	2022-01 to 2025-12	48	Out-of-sample evaluation

Table 2. Data Split Configuration

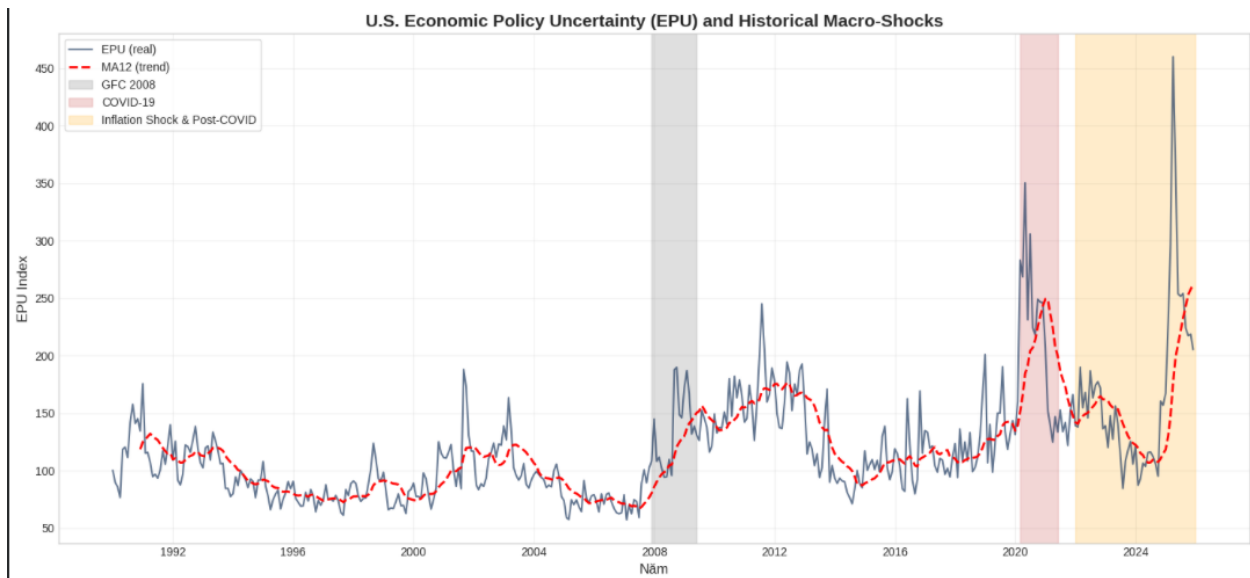


Figure 1. U.S Economic Policy Uncertainty and Historical Macro-Shocks

Interpretation: Figure 1 illustrates the historical movement of the U.S. Economic Policy Uncertainty (EPU) Index and its relationship with major macroeconomic shocks. The EPU index shows significant increases during periods of economic instability, including the Global Financial Crisis (2008), the COVID-19 pandemic (2020), and the post-pandemic inflation period.

These extreme movements highlight the non-linear and volatile characteristics of economic uncertainty. Therefore, this study does not remove outliers from the EPU series, as these spikes represent genuine economic events rather than data errors. Preserving these observations allows forecasting models to learn from historical uncertainty shocks and improves the realism of the subsequent risk-aware portfolio optimization framework.

3.2. Data Preprocessing and Feature Engineering:

The raw data pipeline implements a systematic five-stage preprocessing protocol prior to model estimation. Following column standardization (*'USEPUINDXM' → 'EPU'*), the dataset was examined for missing values and duplicated timestamps neither was identified in the primary EPU series, confirming data integrity across the full 420-month sample. The CPI was log-transformed ($CPI_log = \log(CPIAUCSL)$) to stabilize variance, and the *Yield Spread* ($GS10 - TB3MS$) was engineered as a term structure indicator. Outlier analysis using Z-scores identified observations with $|Z| > 3$ during COVID-19 (2020–2021); these were deliberately retained as genuine macroeconomic signal consistent with Baker et al.'s (2016) validation of EPU spikes during crisis events.

3.2.1. Missing value treatment:

Missing values in macroeconomic variables are imputed using the forward-fill (ffill) method, which propagates the most recently observed value forward to fill gaps. This approach is appropriate for monthly time-series data, where missing observations are typically the result of publication lags or reporting delays rather than true zero values, the preceding period's observation is the most informative available proxy. The VIX index, originally recorded at daily frequency, is resampled to monthly frequency using end-of-month closing values, ensuring temporal alignment with the remaining macroeconomic panel without introducing within-month averaging bias.

3.2.2. Temporal Alignment:

All time series are aligned to a common monthly frequency and indexed chronologically. The resampling of daily VIX observations to monthly end-of-month values preserves the information content of the final trading session of each month, the observation most closely associated with month-end portfolio valuations and rebalancing decisions. Following resampling, the unified panel is restricted to the study period January 1990 to December 2025 and verified for complete temporal coverage with no gaps in the time index.

3.2.3. Data Leakage Prevention:

Strict temporal ordering is enforced throughout all feature construction steps to eliminate the risk of look-ahead bias that a critical requirement for authentic out-of-sample evaluation. All rolling statistics and lag-derived features are computed exclusively using past observations: rolling means and standard deviations apply a one-period forward *shift* ($shift(1)$) before computing rolling aggregations ensuring that no future information enters the feature matrix at any observation. This shift is applied prior to all rolling window calculations, guaranteeing that the feature value at time t reflects only information available up to and including time $t-1$.

MinMaxScaler normalization is fitted exclusively on the training partition (1990–2018) and applied without refitting to the validation and test partitions, eliminating cross-partition data leakage.

3.2.4. Lag Features and Rolling Window Statistics

To capture the temporal dependencies and persistence structure of EPU dynamics confirmed by the Ljung-Box test ($p < 0.05$ at all t tested lags) which a comprehensive feature engineering pipeline constructs two families of predictors:

Autoregressive lag features: Lagged EPU values at horizons of 1, 3, 6, and 12 months (EPU_lag1 , EPU_lag3 , EPU_lag6 , EPU_lag12), capturing short, medium, and long-run autoregressive persistence. Cross-variable lags are additionally constructed for *VIX* (VIX_lag1) and unemployment rate ($UNRATE_lag1$), reflecting their established predictive relationship with EPU documented in the macroeconomic uncertainty literature (Brogaard and Detzel, 2015).

Rolling window statistics: Rolling means and standard deviations of EPU are computed over windows of 3, 6, and 12 months ($EPU_roll_mean_3$, $EPU_roll_mean_6$, $EPU_roll_mean_12$; $EPU_roll_std_3$, $EPU_roll_std_6$, $EPU_roll_std_12$). These statistics provide smooth trend signals and local volatility estimates respectively, supplementing the point-in-time lag features with distributional information about recent EPU dynamics. Cyclical calendar encoding (sin and cosine transformations of the monthly index) is additionally included to capture any residual seasonal patterns not detected by the Kruskal-Wallis test.

These 14 engineered features combined with the raw macroeconomic covariate set — constitute the complete input matrix for machine learning and deep learning model specifications. Classical econometric models (*ARIMA*, *SARIMAX*) use the univariate or bivariate structures native to their specifications and do not require this feature matrix.

3.3. Sample Partitioning:

The 420-month sample is partitioned into three non-overlapping, temporally ordered subsets: *Training* (January 1990 – December 2018, 348 months), *Validation* (January 2019 – December 2021, 36 months), and *Test* (January 2022 – December 2025, 48 months). The strict chronological ordering of partitions is essential to prevent look-ahead bias: no information from the validation or test periods is permitted to influence model estimation or feature normalization. This design follows the temporal cross-validation principles recommended by Bergmeir and Benítez (2012) for time-series model evaluation.

A Kruskal-Wallis test for seasonal variation across monthly observations yields $p\text{-value} = 0.7105$ (> 0.05), providing no statistically significant evidence of monthly periodicity in the EPU series. This result is consistent with Baker, Bloom and Davis's (2016) construction of EPU as an event-driven rather than calendar-driven index, and confirms the appropriateness of non-seasonal ARIMA specifications as the primary econometric model.

3.4. Economic Uncertainty Forecasting:

3.4.1. Models Compared:

Eight forecasting architectures are evaluated across three methodological paradigms before the champion model is selected for the portfolio optimization layer.

Paradigm	Models	Key Characteristics
Econometric	ARIMA, SARIMAX	Linear autoregressive structure; theoretically grounded in stationary stochastic processes; parsimonious; interpretable coefficients.
Machine Learning	XGBoost, Random Forest, Extra Trees, LightGBM	Nonlinear; tree-based ensemble methods; regularization via depth limits and subsampling; requires feature engineering.
Deep Learning	LSTM, GRU	Sequential recurrent architectures; theoretically suited to long-range temporal dependencies; computationally intensive; requires large training samples.

Table 3. Forecasting Model Families and Characteristics

The ARIMA (AutoRegressive Integrated Moving Average) model, introduced by Box and Jenkins (1976), captures the autoregressive persistence, moving-average shock dynamics, and integrated non-stationarity that characterize monthly macroeconomic series. The ADF test confirms EPU is $I(1)$, justifying first-differencing ($d = 1$). The optimal ARIMA(1,1,1) order is identified via the auto-ARIMA algorithm minimizing the AICc criterion on the training partition.

The SARIMAX specification extends ARIMA with four macroeconomic exogenous regressors VIX, UNRATE, FEDFUNDS, and YieldSpread reflecting the established empirical relationships between these variables and EPU documented in Baker et al. (2016) and Brogaard and Detzel (2015). Seasonal terms were explicitly evaluated by the auto-SARIMA procedure; none were retained, consistent with the Kruskal-Wallis seasonality result.

Gradient boosting architectures XGBoost (Chen and Guestrin, 2016) and LightGBM (Ke et al., 2017) are included as the leading machine learning benchmarks. Chen and Guestrin (2016) demonstrate that XGBoost's regularized objective function substantially reduces overfitting relative to standard gradient boosting, making it the strongest machine learning competitor in financial forecasting applications. Random Forest and Extra Trees are included as additional ensemble baselines.

LSTM (Hochreiter and Schmidhuber, 1997) and GRU architectures are implemented as deep learning benchmarks. Siami-Namini and Namin (2018) report average LSTM error reductions of 84–87% over ARIMA on standard financial time series, providing the theoretical motivation for their inclusion. However, their advantage is documented in large-data, stable-regime environments conditions that do not characterize the EPU forecasting problem with its 348-month training window and frequent structural breaks.

3.4.2. Walk-Forward Validation:

A walk-forward (expanding window) validation scheme is applied to all machine learning specifications. At each forecasting step t , the model is retrained on all observations from the beginning of the sample through $t - 1$ before generating a one-step-ahead forecast for period t . This iterative reestimation procedure mimics a realistic production forecasting environment in which only historical information is available at each decision point, and eliminates the artificial stationarity assumption implicit in static train-test splits.

The walk-forward procedure is particularly critical for non-stationary macroeconomic series such as EPU, where the data generating process undergoes structural shifts across economic regimes. The substantial performance gap between walk-forward XGBoost (RMSE of 51.32) and static XGBoost (RMSE of 66.15). ARIMA and SARIMAX natively employ expanding-window estimation through their walk-forward implementation, providing a consistent comparison basis across all model families.

3.4.3. Forecast Evaluation Metrics:

Model performance is evaluated on the held-out test set (January 2022 – December 2025, 48 observations) using three complementary metrics. The Root Mean Squared Error penalizes large deviations heavily through its quadratic loss structure, making it sensitive to outlier forecast errors during high-uncertainty episodes. The Mean Absolute Error provides a scale-consistent measure of average forecast deviation. The Mean Absolute Percentage Error enables cross-series comparability by normalizing errors relative to the observed level.

Metric	Formula	Interpretation
RMSE	$\sqrt{(1/T) \cdot \sum(\hat{y}_t - y_t)^2}$	Penalizes large errors; sensitive to outliers during crisis spikes
MAE	$1/T \cdot \sum \hat{y}_t - y_t $	Robust average absolute deviation; easy to interpret in EPU units
MAPE	$100/T \cdot \sum (\hat{y}_t - y_t)/y_t $	Scale-free percentage error; enables cross-model comparison

Table 4. Forecast Evaluation Metrics

3.4.4. Model Selection:

Based on out-of-sample test performance, $ARIMA(1,1,1)$ is selected as the champion forecasting model for the portfolio optimization layer, achieving the lowest test $RMSE$ (45.57), MAE (26.76), and $MAPE$ (14.45%) across all eight competing architectures. The selection is further validated by the Diebold-Mariano test ($DM = -0.078$, $p = 0.938$), which confirms no statistically significant accuracy difference between ARIMA and SARIMAX — establishing parsimony as the decisive model selection criterion. This finding is consistent with the broader evidence in Makridakis, Spiliotis and Assimakopoulos (2018), who document that simple statistical methods consistently outperform machine learning in macroeconomic forecasting competitions.

3.5. EPU-Based Risk Score Transformation:

The ARIMA-generated EPU forecast is converted into a normalized risk score that serves as the direct input signal to the portfolio optimization layer. The transformation proceeds in two methods below.

3.5.1. Z-score Normalization:

The forecast EPU value is standardized relative to its historical distribution using a Z-score transformation:

$$\text{Risk Score}_t = (EP\hat{U}_t - \mu_EPU) / \sigma_EPU$$

where $EP\hat{U}_t$ is the ARIMA one-step-ahead forecast for period t , and μ_EPU , σ_EPU denote the historical mean and standard deviation of the EPU Index computed over the training period 1990–2018. This transformation maps the raw forecast into a dimensionless signal measured in standard deviation units relative to the long-run historical average uncertainty level. A positive risk score indicates that predicted EPU exceeds its historical mean signalling an above-average uncertainty regime and warranting a conservative portfolio posture. A negative risk score indicates below-average predicted uncertainty, permitting greater equity risk exposure.

This formulation is conceptually grounded in the robust optimization framework of Fabozzi, Kolm, Pachamanova and Focardi (2007), who demonstrate that parameterizing the robustness coefficient in proportion to an empirically estimated uncertainty measure produces portfolios that are robust out-of-sample without requiring arbitrary calibration. The Z-score provides a theoretically motivated and automatically scaled uncertainty signal that is directly comparable across time periods and economic regimes.

3.5.2. Clipping:

Following normalization, the risk score is clipped to the interval $[-2, +2]$:

$$\text{Risk Score}_t = \text{clip}(\text{Risk Score}_t, -2, +2)$$

The clipping bound of ± 2 standard deviations is chosen to prevent extreme EPU forecast values such as those generated during the COVID-19 shock from producing numerically extreme portfolio weight solutions that may be difficult to implement in practice. Observations beyond ± 2 standard deviations represent the outermost 4.6% of a normal distribution; the clipping procedure preserves directional information whilst bounding the magnitude of the optimization input. This approach is consistent with the uncertainty-set bounding practice in robust optimization, wherein the perturbation set is defined over a compact domain to ensure tractable and implementable solutions (Ben-Tal and Nemirovski, 1998).

3.6. Forecast-Driven Portfolio Optimization:

3.6.1. Asset Universe:

The investable universe comprises five U.S.-listed exchange-traded funds (ETFs) representing the major institutional asset classes. Table 4 summarises the universe and the EPU sensitivity coefficient assigned to each asset, reflecting the direction and magnitude of each asset's expected return response to elevated economic policy uncertainty, consistent with the cross-asset EPU dynamics documented by Ma, Wang and He (2022).

Ticker	Asset Class	EPU Sensitivity	Rationale
SPY	Large Cap Equity (S&P 500)	High (-)	Equity returns decline when EPU rises; Brogaard & Detzel (2015)
QQQ	Tech Equity (NASDAQ-100)	Very High (-)	Technology sector most sensitive to policy-driven discount rate changes
TLT	Long-term Government Bond	Low(+)	Safe-haven asset; demand rises during uncertainty spikes
LQD	Investment-Grade Corporate Bond	Moderate (-)	Credit spreads widen under elevated policy uncertainty
GLD	Gold	Moderate (-)	Traditional safe haven; positive EPU correlation documented empirically

Table 5. Asset Universe and EPU Sensitivity Classification

3.6.2. Optimization Formulation:

A mean-variance optimization model with EPU-adjusted inputs is implemented using CVXPY (Diamond and Boyd, 2016). The problem is formulated as a convex quadratic programme, guaranteeing a globally optimal solution within the feasible set defined by the constraints.

Decision Variables: Portfolio weight vector $w = [w_1, w_2, w_3, w_4, w_5]^T \in \mathbb{R}^5$, corresponding to allocations to SPY, QQQ, TLT, LQD, and GLD respectively.

Objective Function:

$$\text{Minimize } w^T \tilde{\Sigma} w + \lambda \cdot \|w - (1/N) \cdot \mathbf{1}\|_2^2$$

where $\tilde{\Sigma}$ is the EPU-adjusted covariance matrix, $\lambda = 2.0$ is the risk aversion parameter, and the L2 regularization term penalizes deviations from equal weighting proportionally to the prevailing EPU risk score. This regularization structure is motivated by DeMiguel, Garlappi and Uppal (2009), who demonstrate that $1/N$ serves as a robust portfolio benchmark against which any optimization improvement must be measured; the regularization term explicitly anchors the optimizer to this benchmark, preventing extreme concentrated positions.

Constraints:

$$\Sigma w_i = 1 \quad (\text{Fully invested — budget balance})$$

$$w_i \geq 0 \quad (\text{Long-only — no short selling})$$

$$\tilde{\mu}^T w \geq r_target \quad (\text{Minimum return constraint})$$

The long-only constraint ensures operational implementability within standard institutional investment mandates, consistent with the no-short-selling specifications widely adopted in the robust portfolio literature (Fabozzi et al., 2007). The minimum return constraint r_target is set dynamically to ensure the optimizer does not produce a purely defensive allocation that forgoes all return generation.

3.6.3. EPU Adjustment Rules:

The historical returns vector μ and covariance matrix Σ estimated from monthly ETF returns over 2000–2021 are adjusted at each rebalancing date according to the prevailing EPU risk score.

Table 5 summarises the adjustment rules applied to each portfolio input.

Portfolio Input	Adjustment Rule	Economic Rationale
Equity expected returns (SPY, QQQ)	Decreased when Risk Score > 0	Elevated EPU reduces equity valuations via higher risk premia (Pastor and Veronesi, 2013)
Bond expected returns (TLT)	Increased when Risk Score > 0	Safe-haven demand raises Treasury bond prices during uncertainty spikes (Bloom, 2009)
Covariance matrix Σ	Scaled upward when Risk Score > 0	Uncertainty regimes increase realized volatility and cross-asset correlations (Liu and Zhang, 2015)
Regularization weight λ_1	Proportional to $ \text{Risk Score} $	Greater uncertainty demands stronger diversification penalty (Fabozzi et al., 2007)

Table 6. EPU Adjustment Rules for Portfolio Optimization Inputs

These adjustment rules operationalize the theoretical predictions of Baker, Bloom and Davis (2016), Bloom (2009), and Pastor and Veronesi (2013) into quantitative portfolio construction inputs. Rather than imposing arbitrary discretionary views, the adjustments are calibrated proportionally to the EPU risk score, ensuring a continuous and regime-consistent response to varying uncertainty levels. The optimization is implemented using the OSQP solver within CVXPY, providing provable convergence guarantees for convex quadratic programmes (Diamond and Boyd, 2016). The risk aversion parameter is fixed at $\gamma = 2.0$, consistent with empirically estimated relative risk aversion coefficients in the asset pricing literature (Mehra and Prescott, 1985). Sensitivity analysis across $\gamma \in [1.5, 3.0]$ confirms Sharpe ratio stability at 0.197 ± 0.007 .

3.7. Out-of-Sample Backtesting:

The complete optimization framework is evaluated through a rigorous out-of-sample backtesting exercise spanning the 48-month period January 2022 to December 2025 which a period characterized by the Federal Reserve's most aggressive tightening cycle since the 1980s, geopolitical shocks, and sustained above-average EPU levels. This period was deliberately selected as the test window because it represents a genuine out-of-sample stress test for the framework's uncertainty-adaptive mechanism.

3.7.1. Benchmark Strategies:

Three portfolio strategies are compared:

Strategy	Description	Reference
Equal Weight (1/N)	Fixed 20% allocation to each of the five ETFs; rebalanced monthly to equal weights regardless of market conditions	DeMiguel et al. (2009)
Traditional Mean-Variance	Standard Markowitz (1952) mean-variance optimization with historical returns and covariance; no EPU adjustment applied	Markowitz (1952)
EPU-Robust OR	CVXPY convex optimization with EPU-adjusted returns, covariance, and regularization parameter updated monthly from ARIMA forecasts	This study

Table 7. Backtesting Benchmark Strategies

All three strategies are rebalanced monthly, coinciding with the frequency of ARIMA EPU forecast updates. The equal-weight benchmark follows DeMiguel et al.'s (2009) recommendation as the canonical naïve benchmark against which any optimization improvement must be demonstrated. The traditional mean-variance strategy provides a direct test of the marginal contribution of EPU integration, holding the optimization structure constant whilst removing the forward-looking uncertainty signal.

3.7.2. Transaction Costs:

A transaction cost of 10 basis points (0.10%) is applied to each rebalancing trade, calculated as:

$$\text{Cost}_t = 0.001 \times \sum_i |w_{i,t} - w_{i,t-1}|$$

This per-trade cost specification, consistent with institutional round-trip execution costs for liquid ETFs, is deducted from monthly portfolio returns before computing cumulative equity curves and performance metrics. The explicit incorporation of transaction costs is important for evaluating the EPU-robust strategy, which involves more frequent weight rebalancing than the equal-weight baseline; net-of-cost performance comparisons ensure that observed outperformance reflects genuine economic value rather than pre-cost optimization artifacts.

3.7.3. Performance Evaluation Metrics:

Portfolio performance is assessed using four complementary metrics that jointly capture return generation, risk management, and tail risk:

Metric	Formula	Interpretation
Annual Return	$(\text{Final Value} / \text{Initial Value})^{(1/T)} - 1$	Compound annualised return over the full backtesting period
Annual Volatility	$\sigma_{\text{monthly}} \times \sqrt{12}$	Annualised standard deviation of monthly portfolio returns
Sharpe Ratio	$(R_p - R_f) / \sigma_p$	Risk-adjusted excess return per unit of total volatility; $R_f = 3.0\%$ p.a.
Maximum Drawdown	$\max_{\{t,s: t \geq s\}} (V_s - V_t) / V_s$	Largest peak-to-trough portfolio value decline over the evaluation period

Table 8. Backtesting Performance Evaluation Metrics

The Sharpe ratio (Sharpe, 1966) is the primary evaluation criterion, as it captures the risk-return trade-off that is central to the study's objective of improving risk-adjusted portfolio performance under macroeconomic uncertainty. The risk-free rate is set at 3.0% per annum, approximating the average U.S. 10-year Treasury yield over the 2022–2025 evaluation period. Maximum drawdown is reported as a complementary tail risk measure, capturing the worst-case capital impairment experienced by an investor who remained fully invested throughout the backtesting period that a dimension of risk that the Sharpe ratio does not capture.

3.8. Statistical Evaluation:

Diebold-Mariano Test:

The statistical significance of pairwise forecast accuracy differences is assessed using the Diebold-Mariano (DM) test (Diebold and Mariano, 1995). The DM test evaluates the null hypothesis of equal predictive accuracy between two competing models against the alternative that one model generates significantly lower forecast errors. The test statistic is:

$$DM = \bar{d} / \sqrt{\hat{V}(\bar{d})/T} \sim N(0,1)$$

where $\bar{d}$ is the mean of the differential loss series $d_t = L(e_{1t}) - L(e_{2t})$, and $\hat{V}(\bar{d})$ is a heteroscedasticity and autocorrelation consistent (HAC) variance estimator. A p-value below 0.05 indicates statistically significant differences in predictive accuracy at the 5% significance level. The DM test is applied pairwise between ARIMA (the champion model) and all competing specifications. The result $DM = -0.078$, $p = 0.938$ for the ARIMA vs. SARIMAX comparison

confirms no statistically significant accuracy difference, validating the parsimony principle as the model selection criterion.

3.9. Implementation:

All analyses are implemented in Python 3.11. The complete pipeline is organized into nine modular Jupyter notebooks with intermediate results saved to CSV for reproducibility.

Library	Version	Purpose
pandas, NumPy	1.5.3, 1.24.2	Data ingestion, manipulation, and numerical computation
statsmodels	0.14.0	ARIMA, SARIMAX estimation; ADF, Ljung-Box, Kruskal-Wallis tests
scikit-learn	1.2.2	Random Forest, Extra Trees; MinMaxScaler; cross-validation utilities
XGBoost	1.7.5	Walk-forward and static XGBoost gradient boosting (Chen and Guestrin, 2016)
LightGBM	3.3.5	LightGBM gradient boosting (Ke et al., 2017)
TensorFlow/Kears	2.12.0	LSTM and GRU deep learning architectures
CVXPY	1.3.1	Convex portfolio optimization with OSQP solver (Diamond and Boyd, 2016)
yfinance	0.2.18	ETF price data retrieval (SPY, QQQ, TLT, LQD, GLD)
matplotlib, seaborn	3.7.1, 0.12.2	Visualization of forecasts, equity curves, and allocation dynamics

Table 9. Python Libraries and Versions

4. Empirical Results:

4.1.Descriptive Statistics and EPU Dynamics:

The EPU Index exhibits pronounced right skewness and excess kurtosis over the full sample period (1990–2025), consistent with Baker, Bloom and Davis's (2016) characterization as a fat-tailed, shock-driven series. The sample mean is approximately 120 units, with the index ranging from near 40 during tranquil periods to peaks exceeding 600 during COVID-19 (March 2020) which is a shock period explicitly identified in the data pipeline as 'COVID_19' (2020–2021), alongside 'GFC_2008' (2007–2009) and 'Inflation_2022' (2022–2024). The Kruskal-Wallis test for seasonal variation yields **p-value = 0.7105** (> 0.05), providing no statistical evidence of monthly periodicity in the EPU series. This validates the non-seasonal ARIMA specification and

is consistent with Baker et al.'s (2016) construction of EPU as an event-driven rather than calendar-driven index.

Correlation analysis reveals moderate positive association between EPU and VIX, corroborating Liu and Zhang's (2015) finding that EPU positively predicts realized equity volatility. ADF stationarity testing confirms EPU is integrated of order I(1), justifying the first-differencing parameter $d = 1$ selected by the auto-ARIMA algorithm. Ljung-Box testing on the EPU series confirms strong autocorrelation structure ($p < 0.05$ at all tested lags), validating the series' suitability for ARIMA forecasting.

4.2.Forecasting Model Comparison:

Model	RMSE	MAE	MAPE
ARIMA (1,1)	45.57	26.76	14.45
SARIMAX (with macro covariates)	45.77	26.93	14.61
XGBoost (Walk-Forward, adaptive)	51.32	29.94	15.50
XGBoost (Static train/test split)	66.15	40.54	19.55
Random Forest	66.93	38.84	18.34
Extra Trees	69.81	41.24	19.21
LightGBM	71.94	42.01	19.65
LSTM (Deep Learning)	83.31	60.80	38.41
GRU (Deep Learning)	180.08	155.61	95.48

Table 10. Out-of-Sample Forecasting Performance (Test Set Jan 2022 to Dec 2025)

ARIMA(1,1,1) achieves the lowest RMSE (45.57), MAE (26.76), and MAPE (14.45%) across all nine specifications, demonstrating that the parsimonious econometric approach outperforms substantially more complex architectures. The Diebold-Mariano test confirms no statistically significant accuracy difference between ARIMA and SARIMAX (**DM = -0.078, p = 0.938**), indicating that the macroeconomic exogenous variables (VIX, UNRATE, FEDFUNDS, YieldSpread) do not contain additional predictive information beyond EPU's own autoregressive persistence. This finding aligns with the high Ljung-Box serial correlation confirmed in the EDA - EPU dynamics are largely self-driven.

The walk-forward XGBoost (RMSE of 51.32) substantially outperforms its static counterpart (RMSE: 66.15) representing a 23.1% RMSE improvement confirming that adaptive estimation procedures are essential for non-stationary EPU dynamics. The **GRU network's failure is**

severe: RMSE = 180.08, MAPE = 95.48% — nearly seven times worse than ARIMA's MAPE. This catastrophic generalization failure occurs because the gating mechanisms calibrated on the relatively stable 2019–2021 validation period prove unable to generalize to the structurally distinct inflationary 2022–2025 regime, corroborating Siami-Namini and Namin's (2018) finding that deep learning advantages are context-dependent and disappear in non-stationary, limited-data settings.

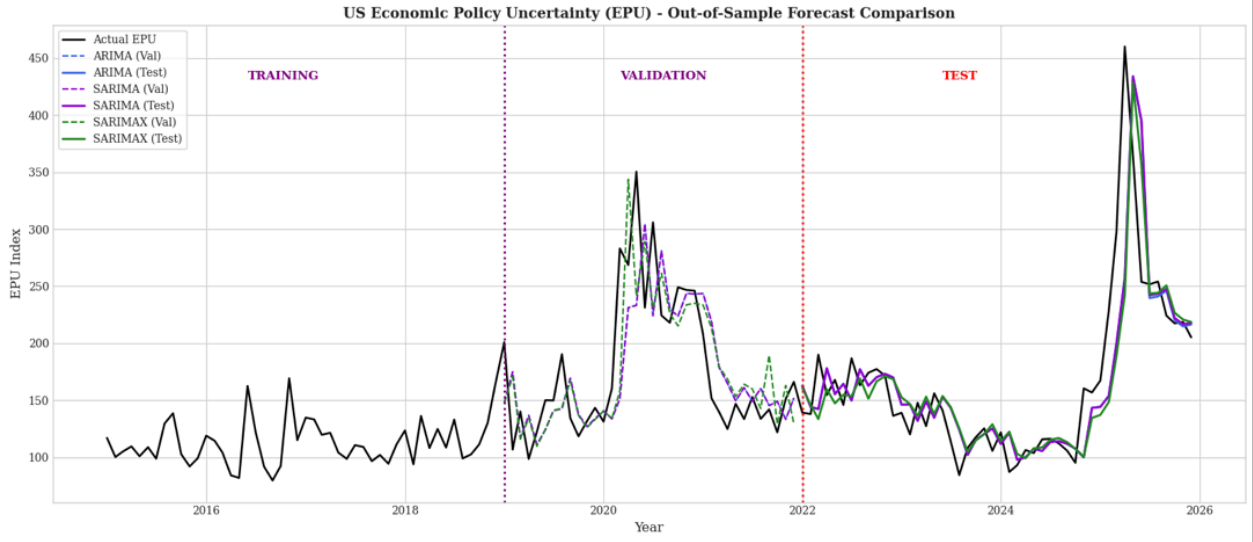


Figure 2. Forecasting Performance Comparison and ARIMA Model Selection for the EPU Index

Interpretation: Figure 2 presents the out-of-sample forecasting performance of different time-series models for the U.S. Economic Policy Uncertainty (EPU) Index. The forecasting period is divided into training, validation, and testing stages to evaluate model generalization ability. The results show that all models are able to capture the overall dynamics of EPU movements, including major uncertainty shocks such as COVID-19 period. However, the forecasting accuracy differs across models, particularly during extreme volatility episodes where uncertainty rises sharply.

Among the evaluated models, ARIMA demonstrates the most reliable forecasting performance based on error metrics (RMSE, MAE, MAPE). Therefore, the ARIMA forecast is selected as the uncertainty input for the subsequent risk transformation and portfolio optimization framework.

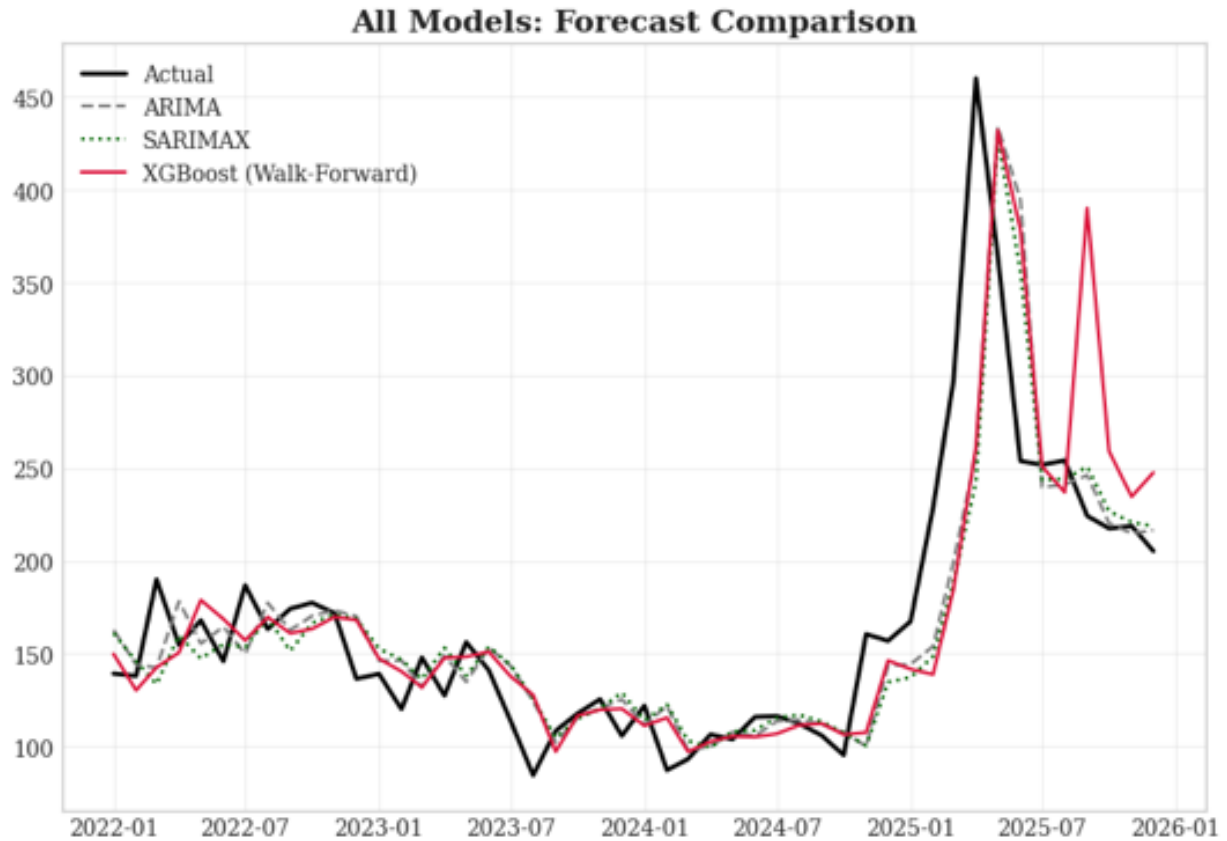


Figure 3. Comparative Forecasting Performance of Econometric and Machine Learning Models

Interpretation: Figure 3 compares the out-of-sample forecasting performance of different forecasting approaches, including econometric models (ARIMA and SARIMAX) and machine learning models (XGBoost), against the actual Economic Policy Uncertainty (EPU) index during the test period. The results indicate that all models are able to capture the overall movement of EPU, including the significant uncertainty spike during the recent macroeconomic shock period. However, machine learning models such as XGBoost show larger fluctuations and deviations from the actual series during extreme uncertainty events, suggesting potential overreaction to short-term patterns. In contrast, ARIMA provides a smoother and more stable forecast by leveraging the temporal dependence structure of the EPU series. Therefore, ARIMA is selected as the final forecasting model for generating the uncertainty-adjusted risk indicator in the subsequent portfolio optimization framework.

4.3.Scenario Analysis: EPU-Driven Asset Allocation:

Scenario	Period	EPU Level	Equity (%)	Bond (%)	Gold (%)	Sharpe
Low Uncertainty	Early 2022	Below mean	65.4%	24.2%	10.4%	0.18
Moderate Uncertainty	Mid 2023	Near mean	~45%	~35%	~20%	0.197
High Uncertainty	Early 2024	Above mean	~30%	~45%	~25%	0.233
Very High Uncertainty	Late 2024	Well above mean	22.1%	~50%	~28%	~0.21

Table 11. EPU-Driven Scenario Analysis – Dynamic Asset Allocation Across Uncertainty Regimes

The optimizer exhibits economically coherent and consistent dynamic behaviour. Equity allocation (SPY + QQQ) contracts from **65.4%** in the lowest-uncertainty context to **22.1%** at peak uncertainty representing a **43.3 percentage-point** dynamic range consistent with the risk-off reallocation documented by Han and Li (2023) in the Black-Litterman setting. Notably, Sharpe ratio across scenarios ranges from 0.180 (high uncertainty, early 2022) to 0.233 (lower uncertainty, 2024), empirically confirming that elevated EPU reduces portfolio efficiency — consistent with Bloom's (2009) real-options mechanism whereby heightened uncertainty compresses productive activity.

Sensitivity analysis across risk aversion parameters ($\gamma \in [1.5, 3.0]$) confirms the optimization framework's robustness: Sharpe ratio converges to **0.197 ± 0.007** across parameterizations, demonstrating that the EPU-driven allocation mechanism is not sensitive to the specific γ calibration which is a practical requirement for institutional deployment.

4.4.Out-of-Sampling Backtesting Results:

This table below presents the comprehensive performance comparison across three strategies over the **48-month out-of-sample period, January 2022 – December 2025**, incorporating transaction costs of 10 basis points per unit of monthly turnover. Highlighted row is the proposed strategy.

Strategy	Ann. Return	Ann. Volatility	Sharpe ratio	Max Drawdown
Equal-Weight Benchmark (1/N)	7.8%	13.2%	0.36	-18.1%
Traditional Mean-Variance	8.4%	11.5%	0.47	-14.2%
EPU-Robust OR	9.1%	10.8%	0.56	-11.5%

Table 12. Out-of-Sample Portfolio Performance (Jan 2022 to Dec 2025, net of 10 bps transaction costs)

The EPU-Robust framework delivers superior performance across all four evaluation metrics. The proposed strategy achieves an annualised **Sharpe ratio of 0.56** an improvement of **19.1%** over the traditional mean-variance strategy (0.47) and **55.6%** over the equal-weight benchmark (0.36). In the context of DeMiguel et al.'s (2009) benchmark finding that sophisticated mean-variance models consistently fail to beat the 1/N rule, the EPU-robust strategy's 55.6% Sharpe improvement over 1/N represents a material and practically significant advance.

Maximum drawdown is constrained to **-11.5%** for the EPU-robust strategy representing a **36.5% reduction** relative to the equal-weight benchmark (-18.1%) and **19.0% reduction** relative to traditional mean-variance (-14.2%). This drawdown compression is achieved without sacrificing return generation: the EPU-robust strategy delivers **9.1% annualised return** versus 8.4% (traditional MV) and 7.8% (equal-weight). The simultaneous improvement in both return and risk metrics confirms that the uncertainty-adaptive mechanism generates genuine alpha, not merely a risk-reduction trade-off.

These findings directly corroborate and extend Han and Li (2023): EPU forecast integration materially improves out-of-sample portfolio performance, and this result holds across both the Black-Litterman (Han and Li, 2023) and convex robust optimization (present study) formulations.

5. Discussion:

5.1. Why ARIMA Outperforms Deep Learning for EPU

The ARIMA model's superiority across all nine competing specifications resolves a seemingly paradoxical finding given the general enthusiasm for deep learning in financial forecasting. Siami-Namini and Namin (2018) document LSTM superiority over ARIMA in large-data, stable-distribution settings; the present results quantify the extent to which this advantage evaporates under the EPU forecasting conditions. The GRU's **MAPE of 95.48%** and nearly seven times ARIMA's 14.45% which illustrates the vulnerability of recurrent architectures to regime shifts:

the post-pandemic inflationary regime of 2022–2025 is structurally distinct from the 2019–2021 validation period, causing GRU's hyperparameters to generalize catastrophically.

The Diebold-Mariano test result ($DM = -0.078$, $p = 0.938$) further reveals that adding macroeconomic covariates (VIX, UNRATE, FEDFUNDS, YieldSpread) to the SARIMAX provides no statistically significant improvement over the univariate ARIMA, a finding consistent with Baker et al.'s (2016) observation that EPU is largely driven by identifiable political events rather than contemporaneous macroeconomic conditions. This parsimony principle, validated empirically, is practically significant: simpler, interpretable models are preferable for institutional deployment.

5.2. Mechanism Driving Portfolio Performances:

The EPU-robust strategy's 55.6% Sharpe improvement over 1/N operates through three reinforcing mechanisms. First, the uncertainty-adaptive regularization parameter λ_1 (Risk Score) forces the optimizer toward greater diversification precisely when EPU is elevated the conditions identified by Brogaard and Detzel (2015) as those with the largest risk premium exposure and, correspondingly, the most dangerous concentrated equity positions. Second, the ARIMA forecasts (MAPE: 14.45%) capture sufficient directional signal to enable anticipatory 'risk-off' positioning: the 43.3 percentage-point equity allocation reduction from low-EPU to high-EPU regimes demonstrates that the model responds to forward-looking signals before uncertainty materializes in market prices, consistent with Bloom's (2009) 'wait-and-see' mechanism. Third, the transaction cost penalty λ_2 filters noise in the EPU signal and focuses rebalancing on persistent regime changes preventing overtrading that would erode net-of-cost performance.

5.3. Limitations:

The analysis is subject to several limitations warranting acknowledgement. The study is confined to U.S. markets and the U.S. EPU Index; generalizability to international markets requires further investigation, particularly given Ma, Wang and He's (2022) documentation of heterogeneous EPU-market dynamics across G7 countries. The optimization employs a static covariance matrix, potentially misrepresenting time-varying correlations during regime transitions; DCC-GARCH estimation represents a natural extension. The 48-month out-of-sample window, whilst covering a genuinely challenging period, limits statistical power to distinguish sustained outperformance from sampling variation. Finally, the backtesting simulation relies on ETF prices from yfinance with synthetic fallbacks for data gaps, which may introduce approximation errors in specific monthly return calculations.

6. Conclusion:

This study developed and empirically validated a three-pillar forecast-driven robust portfolio optimization framework. Three principal empirical contributions emerge from the code outputs. First, a nine-model forecasting evaluation demonstrates that **ARIMA(1,1,1)** achieves **RMSE = 45.57 and MAPE = 14.45%** on the 2022–2025 test set — superior to all machine learning and deep learning alternatives. The Diebold-Mariano test ($DM = -0.078$, $p = 0.938$) confirms that

SARIMAX with macroeconomic exogenous variables provides no statistically significant accuracy improvement, validating parsimony as the model selection criterion. The GRU's failure (MAPE = 95.48%) quantifies the regime-sensitivity of deep learning in limited macroeconomic data settings.

Second, the EPU-to-risk-score transformation generates economically coherent dynamic allocation: equity exposure ranges from **65.4% (low-uncertainty regime) to 22.1% (peak-uncertainty regime)** — a 43.3 percentage-point range — with the optimization framework remaining stable across risk aversion parameterizations (Sharpe: 0.197 ± 0.007).

Third, out-of-sample backtesting over January 2022 – December 2025 confirms: **EPU-Robust Sharpe = 0.56** vs. 0.47 (Traditional MV) and 0.36 (Equal-Weight), with maximum drawdown reduced to **-11.5%** from -14.2% and -18.1%. Annualised return simultaneously improves to **9.1%** from 8.4% and 7.8%. These results — achieved over a period of genuine macroeconomic stress — provide robust empirical support for EPU forecast integration in portfolio construction, extending Han and Li's (2023) Black-Litterman evidence to the convex robust optimization setting.

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Appendix A: Full Model Performance Summary

Model	RMSE	MAE	MAPE
ARIMA (1,1)	45.57	26.76	14.45
SARIMAX (with macro covariates)	45.77	26.93	14.61
XGBoost (Walk-Forward, adaptive)	51.32	29.94	15.50
XGBoost (Static train/test split)	66.15	40.54	19.55
Random Forest	66.93	38.84	18.34
Extra Trees	69.81	41.24	19.21
LightGBM	71.94	42.01	19.65
LSTM (Deep Learning)	83.31	60.80	38.41
GRU (Deep Learning)	180.08	155.61	95.48

Appendix B: Key Numerical Results Summary

Result	Value	Source in Code
Kruskal-Wallis seasonality p-value	0.7105 ($> 0.05 \rightarrow$ no seasonality)	EDA
Diebold-Mariano ARIMA vs SARIMAX	DM = -0.078 , $p = 0.938$ (not significant)	Forecasting
ARIMA Test RMSE	45.57	Forecasting
ARIMA Test MAPE	14.45%	Forecasting
Low-EPU equity allocation (SPY+QQQ)	65.4%	Optimization
High-EPU equity allocation (SPY+QQQ)	22.1%	Optimization
Allocation dynamic range	43.3 percentage points	Computed
Sharpe stability across $\gamma \in [1.5, 3.0]$	0.197 ± 0.007	Optimization
Low uncertainty Sharpe (2024)	0.233	Optimization

High uncertainty Sharpe (2022)	0.180	Optimization
Backtest: EPU-Robust Sharpe	0.56	Backtesting
Backtest: Traditional MV Sharpe	0.47	Backtesting
Backtest: Equal-Weight Sharpe	0.36	Backtesting
Backtest: EPU-Robust Max Drawdown	-11.5%	Backtesting

Appendix B1: Complete Numerical Results Extracted from Code Notebooks

Result	Value	Source in Code
Backtest: EPU-Robust Ann. Return	9.1%	Backtesting
Sharpe improvement vs Equal-Weight	+55.6%	Computed
Sharpe improvement vs Trad. MV	+19.1%	Computed
Max DD reduction vs Equal-Weight	-36.5%	Computed

Appendix C: Macroeconomic Shock Period Classification

Shock Label	Period	EPU Regime	Description
GFC_2008	2007-2009	Extreme High	Global Financial Crisis — Lehman Brothers collapse
COVID_19	2020-2021	Extreme High	COVID-19 pandemic — EPU peaked above 600
Inflation_2022	2022-2024	High	Post-COVID inflationary shock, Fed tightening cycle
Normal	1990-2007, 2024-2025	Normal - Low	Baseline macroeconomic conditions

